Tasks - Business Case–Driven ML Pipeline Report

You will produce a short technical report + notebook that demonstrates a complete machine learning workflow on a dataset relevant to your major.

Requirements

Mandatory deliverables

- Final report must be **submitted to LSM** as a `.pdf` file, with all results visible in the output cells and all explanations written in code/text cells. You need to have ready ipynb file for your defense in your laptop.
- **Kaggle dataset** (provide the link)
- Clear section headings (**Markdown**)
- **Clean formatting:** code, plots, and tables aligned and readable
- **Leakage-free** preprocessing (fit only on train)
- **File naming convention:** SE-20**_assignment1_Name_Surname (e.g., SE-2023_assignment1_John_Smith).

Submission quality

- The whole report must be **easy to read**, with **logical flow** and **clear explanations**
- All attachments (code, graphs, tables) must be **well formatted and aligned**

Report structure

Important: Every section listed below must appear in your report.
The items under "Examples" are guidance, not strict requirements. Your exact steps depend on your dataset and task. Your main goal is to build a complete ML pipeline and explain the purpose of each step. Use text cells to write the report text.

Dataset selection

Goal: Select a Kaggle dataset related to your major and briefly explain why you chose it.
Mandatory: provide the link.

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Start coding or generate with AI.
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Business understanding

Goal: Explain the real-world task and how ML supports decision-making in your domain.

Examples:

- What decision will the model support?
- Who benefits (stakeholders/users)?
- Expected impact (time/cost/risk/quality/safety)
- Success criteria: metrics + target thresholds (business acceptance targets)

References:

- https://developers.google.com/machine-learning/problem-framing

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Exploratory Data Analysis

Goal: Describe the dataset and extract insights using statistics and plots.

Examples:

- Column/feature descriptions (short data dictionary)
- Summary statistics (mean/median/std; class balance if classification)
- Visualizations (distributions, relationships to target)
- Correlation analysis

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Data cleaning

Goal: Identify and address data quality issues with “before → after” evidence.

Examples:

- Missing values (detect + handle + justify)
- Duplicates (detect + handle)
- Errors/anomalies (invalid values, impossible ranges, inconsistent categories)

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Feature engineering

Goal: Improve feature representation and justify decisions in domain terms.

Examples:

- Feature selection (drop/keep with justification)
- Feature creation: 1–3 new features + domain explanation

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Data preprocessing

Goal: Prepare data for modeling with correct splitting and no data leakage.
Mandatory: Train/test split and leakage-free preprocessing.

Examples:

- Encoding categorical variables
- Scaling numeric features (when needed)
- Pipelines for reproducibility (optional)

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Training

Goal: Train baseline models and briefly justify them.

Required baselines:

- For Regression: Linear Regression + Decision Tree Regressor
- For Classification: Logistic Regression + Decision Tree Classifier

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Evaluation

Goal: Evaluate models with appropriate metrics and connect results to the success criteria.

Examples (guidance):

- For Regression: MAE, MSE/RMSE (optional R^2)
- For Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix
- Comparison table across models + short interpretation

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Conclusion

Goal: Summarize what you learned and whether the result satisfies business requirements.

Examples:

- Key findings + limitations
- Best baseline model + why it likely performed best
- Meets success criteria or not (explain)
- Next steps (data/features/models/validation)

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